

CAPE SAGRO, France

WMO: 077910

Lat: 42.798N

Lon: 9.489E

Elev: 394

StdP: 14.49

Time Zone: 1.00 (EUC)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	38.8	41.1	21.1	16.0	44.3	24.8	19.2	45.5	61.3	60.0	55.6	56.3	10.1	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	84.6	72.8	83.0	72.7	81.7	72.2	76.7	81.8	75.6	80.6	74.6	79.5	7.0	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	133.3	80.5	73.8	128.0	79.6	72.8	123.4	78.6	40.4	82.1	39.3	81.0	38.4	79.6	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 47.1	34.1	26.2	DB	34.7	87.7	3.8	1.5	32.0	88.8	29.8	89.6	27.6	90.5	24.9	91.6	(4)
(5)			WB	31.5	78.1	2.7	1.8	29.5	79.4	28.0	80.4	26.5	81.4	24.5	82.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.3	50.7	50.4	52.9	57.7	64.4	69.4	76.7	76.9	71.7	65.8	57.7	52.6	(6)	
	DBStd	10.31	4.24	4.84	4.75	3.25	4.44	3.83	2.99	2.64	3.65	3.87	4.62	4.78	(7)	
	HDD50	158	43	50	30	1	0	0	0	0	0	0	5	29	(8)	
	HDD65	2166	443	410	376	220	63	7	0	0	3	36	222	386	(9)	
	CDD50	4643	65	60	119	231	445	581	828	833	651	488	234	108	(10)	
	CDD65	1180	0	0	0	1	43	138	363	368	205	60	2	0	(11)	
	CDH74	5678	0	0	0	0	71	293	2420	2338	542	14	0	0	(12)	
	CDH80	708	0	0	0	0	5	13	363	305	22	0	0	0	(13)	
(14) Wind	WSAvg	8.5	8.7	8.9	9.2	8.8	6.9	7.6	6.6	7.0	7.5	9.8	10.8	10.5	(14)	
(15) Precipitation	PrecAvg	24.5	2.1	1.9	2.2	2.2	1.5	1.1	0.4	0.7	2.1	3.3	4.3	2.7	(15)	
	PrecMax	34.4	6.3	5.9	5.7	4.6	5.0	3.7	2.2	2.6	6.1	5.8	9.1	5.4	(16)	
	PrecMin	16.9	0.3	0.0	0.2	0.3	0.2	0.1	0.0	0.0	0.0	0.4	0.7	0.3	(17)	
	PrecStd	4.8	1.3	1.4	1.4	1.1	1.3	0.9	0.6	0.6	1.8	1.7	2.3	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	63.8	65.2	69.6	79.5	82.0	87.3	86.6	82.3	75.4	69.3	67.5	(19)	
		MCWB	53.1	51.7	52.7	60.3	66.9	71.4	70.7	73.0	71.3	65.8	62.2	56.3	(20)	
	2%	DB	59.2	61.1	62.3	67.0	75.4	79.4	85.3	84.4	80.1	73.7	66.6	62.5	(21)	
		MCWB	52.0	51.5	53.2	59.5	66.6	70.7	72.6	73.0	70.0	65.9	61.1	55.3	(22)	
	5%	DB	57.2	58.4	60.3	64.8	72.9	77.4	83.7	82.9	78.7	72.0	64.7	60.1	(23)	
		MCWB	51.5	51.5	53.1	57.2	64.8	70.0	72.9	72.8	69.8	65.4	58.9	54.5	(24)	
	10%	DB	56.0	56.4	58.6	62.8	70.8	75.4	82.1	81.8	77.2	70.4	63.2	58.6	(25)	
		MCWB	50.8	50.4	52.5	56.4	63.4	68.9	72.5	72.4	68.9	64.7	57.8	53.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.1	56.2	57.3	64.5	71.0	75.6	78.3	78.3	76.0	70.6	65.2	59.0	(27)	
		MCDB	59.1	59.7	59.8	67.9	74.7	79.8	83.3	83.4	79.3	73.1	67.7	62.4	(28)	
	2%	WB	54.4	54.0	56.0	61.1	68.7	73.5	76.9	76.7	73.8	68.9	62.6	57.5	(29)	
		MCDB	57.3	57.7	58.4	65.2	73.8	77.3	82.1	82.1	77.6	71.4	65.9	60.8	(30)	
	5%	WB	53.2	52.8	55.3	59.0	66.6	71.8	75.8	75.6	72.3	67.7	60.6	56.0	(31)	
		MCDB	56.2	56.4	57.7	62.8	71.7	76.0	80.9	81.0	76.3	70.4	63.7	59.4	(32)	
	10%	WB	51.9	51.6	54.4	57.7	64.9	70.1	74.8	74.6	70.8	66.3	59.0	54.0	(33)	
		MCDB	55.3	55.3	57.3	61.4	69.3	74.4	79.7	79.9	75.0	69.5	62.4	57.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	7.0	7.0	7.2	7.9	8.0	8.4	8.5	8.3	7.0	6.4	6.4	(35)	
		MCDBR	7.8	8.6	8.9	9.8	10.4	9.3	9.8	9.6	9.3	8.0	7.0	8.2	(36)	
	5% WB	MCWBR	6.1	6.9	7.2	7.0	7.8	7.0	7.4	7.4	6.8	6.3	6.3	6.6	(37)	
		MCWBR	6.6	7.2	6.7	8.0	8.9	8.4	8.5	8.6	8.4	6.4	5.8	8.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.315	0.330	0.356	0.394	0.403	0.401	0.398	0.396	0.385	0.364	0.342	0.316	(40)	
		taud	2.547	2.503	2.446	2.345	2.364	2.407	2.412	2.432	2.450	2.510	2.521	2.550	(41)	
	Edn at Noon		264	277	281	277	277	277	276	273	268	261	250	252	(42)	
		Edh at Noon	22	27	32	38	38	37	36	34	31	26	23	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		533	812	1208	1630	1977	2233	2275	1982	1490	974	597	469	(44)	
	RadStd		49	81	130	97	146	111	94	93	66	85	56	53	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	N/A

Nomenclature: See separate page